Spotify Data Analysis

Data Science 1130: Final Project Report
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Overall Research Topic: Factors Affecting the popularity and the listener engagement of other tracks when more highly popular tracks are present in the album in Spotify; Factors influencing how solo and collaborative tracks differ musically.

Subtopic 1- Impact on Less Popular Tracks in Albums on Spotify

Main Problem: The problem we are trying to target here is how less popular tracks are impacted if they are put together with highly popular tracks in the same album. To answer this problem, we will start off by comparing two albums: one with highly popular tracks and one without highly popular tracks and answering the following question:

"Is the average popularity of other tracks on albums with highly popular tracks higher than that of albums without such tracks?"

Background/Introduction of Problem

This question explores whether the success of a highly popular track can influence the popularity of other tracks on the same album. The hypothesis is that albums containing at least one highly popular track might exhibit higher average popularity for the other tracks, as listening to the popular tracks gives more engagement to the tracks in the album as well as people won't usually change the album as frequently and will go on with the tracks in the album.

Data Analysis Steps

1. **Dataset and Preprocessing**:
    - A Spotify dataset.csv containing information regarding key variables like track_genre, track popularity, track ID are quite useful for our analysis.
    - There were 3 missing values found which were in album_name, artist and track_name columns, it seems it is impossible to find these values as we couldn't possibly find values of a artist who has made a track as there would be number of possibilities and since this values are very small in number we could possibly drop this values. It came to our team's attention that this would be a only efficient way to deal with these missing values.
    - Data filtered, we saw many rows having same values but different genre, this all rows can be combined in just one rows where the genre can be separated by ', ' so now our dataframe will have just one row instead of duplication of rows consisting of multiple genre/whichever genre the track consist of.
2. **Defining a Highly Popular Track**:

    - A highly popular track was defined as having a popularity score greater than 80 (threshold determined based on data distribution) this give us a value of 49 note this value is processed after the data filtration part.
    - So now whichever tracks having popularity greater than 49 are highly popular tracks and other tracks which are below or equal to popularity of 49 are considered as less popular tracks.
3. **Grouping Tracks by Albums**:
    - Tracks were grouped by their track popularity, and albums were categorized into two groups:
        - Albums with at least one highly popular track.
        - Albums without any highly popular tracks.
4. **Calculating Average Popularity**:
    - Now in order to find the average popularity of tracks without highly popular tracks in the album is easy but in order to find the average popularity of tracks in albums with highly popular tracks is somewhat hard to process and is something that we were stuck on but this can be done by mapping the album set for highly popular tracks, we will map on the basis of there album_name and will find this album name in our original data filtered df, note this album names will be decided from the highly popular album set (df) and this tracks will be then placed in our sorted_df1 and the mean for this albums will be calculated and stored in final_mean1 and then we will test final_mean1 vs final_mean2.
5. **Hypothesis Testing**:
    - A t-test is conducted by us to compare the mean popularity of tracks between the two groups of albums, testing whether there is a statistically significant difference.

Main Results/Conclusions

Key Findings:

1. **Average Popularity Comparison:**
    - Albums with at least one highly popular track (popularity > 80) showed a **higher average popularity** for their other tracks compared to albums without any highly popular tracks.
    - The calculated mean popularity for tracks on albums with highly popular tracks (μ1\mu_1μ1) was 36.688382045574635, while the mean popularity for tracks on albums without highly popular tracks (μ2\mu_2μ2) was 24.771246657356848.
2. **Hypothesis Testing Results:**
    - The **t-test** resulted in a p-value of **0.0**, which is less than the significance level (α/alpha = 0.05). This indicates that the difference in means between the two groups is statistically significant.
    - The test supports the **alternative hypothesis (H1)**, concluding that the average popularity of other tracks on albums with highly popular tracks is **higher** than the average popularity of tracks on albums without highly popular tracks.

Drawbacks and Limitations

- **Threshold Selection**: The cutoff of 80 for defining highly popular tracks may be subjective. A sensitivity analysis could provide further insights.
- **Limited Context on Popularity Metric:** Spotify's popularity metric is dynamic and influenced by external factors like marketing campaigns, playlist inclusion, and recentness of release, which were not accounted for in the analysis.
- **Data Quality, Missing Values:** While dropping rows with missing values was efficient, it could lead to a slight loss of valuable information, especially if those tracks had unique characteristics.

Subtopic 2: "What are the differences between solo and collaborative projects in terms of musical qualities?"

Background/Introduction of Problem

This question investigates whether solo and collaborative projects differ in musical qualities such as energy, valence, danceability, and instrumentalness versus group music production.

Data Analysis Steps

1. **Dataset and Preprocessing**:
    - In order to answer our question we need to find out which tracks are made by sole artist and which are collaborative tracks and we will then divide this into two set of df's. We will first need to separate out the artist in order to find how many artist were required to produce a track, this can be done by .apply() method. So, now the tracks are categorized as solo projects (single artist) or collaborative projects (multiple artists).
    - In order to visualize the difference in more depth our team carried out multiple graphs representation and one of the requirements for duration vs popularity for both our sets of solo and collaborative tracks requires duration which is given to us in ms so we converted it to minutes as this would be easier to visualize.
2. **Comparison between Solo and Collaborative Tracks**:
    - Firstly we will proceed with the average values of all musical qualities for both the sets of Solo and collaborative tracks. To have a overlook on the results of musical qualities, we see that danceability, energy, speechiness, acousticness, valence all have almost the same average. Whereas loudness and popularity have some difference in there average.
3. **Visualization**:
    - Barplot is created for showing the comparison of musical qualities between solo tracks and collaborative tracks.
    - Boxplots, Barplot, heatmap, scatterplot and histograms were created for each musical quality, comparing solo and collaborative projects.
    - **Solo Tracks:**

        i. The scatterplot and regression line for solo tracks illustrate the relationship between **danceability** vs **popularity**.
        ii. The **positive slope** suggests that as danceability increases, the popularity of solo tracks also tends to increase. However, the strength of the correlation may vary, depending on the spread of the data points around the line.
    - **Collaborative Tracks:**
        i. Similarly, the scatterplot and regression line for collaborative tracks indicate the relationship between danceability and popularity.
        ii. If the **slope** is steeper for collaborative tracks compared to solo tracks, it implies a stronger relationship between danceability and popularity in collaborative projects. Conversely, a smaller slope would suggest a weaker relationship.
    - **Slope Comparison:**
        i. If the slope for collaborative tracks is higher, it means collaborative projects are more sensitive to changes in danceability when predicting popularity compared to solo tracks.
        ii. A similar or lower slope for collaborative tracks would indicate that the influence of danceability on popularity is comparable or less significant than in solo tracks.
    - **Intercept Comparison:**
        i. The intercept reflects the baseline popularity when danceability is at zero. Differences in intercepts between solo and collaborative tracks could indicate varying baseline popularity levels, potentially due to the inherent appeal of collaborations.
    - The steeper slope (**Collaborative** 6.39 vs. **Solo** 2.63) for collaborative tracks suggests that **danceability** plays a more significant role in driving the popularity of collaborative tracks compared to solo tracks.
    - **collaborative tracks** (slope = 9.96) compared to **solo tracks** (slope = 2.30), suggesting that energy is a more influential factor in the success of collaborative projects, similarly it goes for all the musical qualities .

Main Results/Conclusions

- Collaborative tracks often have different dynamics due to the combined fan bases, creative input, and market appeal, which might explain why their slope or intercept could differ from solo tracks.
- Solo tracks might rely more on the unique style and appeal of the individual artist, making their popularity less tied to danceability or other musical qualities compared to collaborative projects.
- **Energy**: Collaborative projects exhibited higher energy scores on average than solo projects.
- **Danceability**: No significant difference was observed in danceability.

Collaborative tracks exhibit a stronger relationship between musical qualities like **energy** and **danceability** and their popularity, with higher energy and a more pronounced impact of danceability on success compared to solo tracks. Solo projects, in contrast, are more

influenced by the unique style of the individual artist, with popularity less tied to variations in musical qualities.

Drawbacks and Limitations

- The analysis may be influenced by the genre of music, as different genres prioritize varying musical qualities. Solo and collaborative projects within specific genres could behave differently, skewing the results.
- Factors like marketing, artist popularity, and audience engagement (e.g., social media influence) were not considered but could significantly impact track popularity, overshadowing the influence of musical qualities.

Sources:

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